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RESEARCH ARTICLE

Assessing Ergonomic Risk Loads on Pilots Across Diverse Aircraft Types

Uçak Tiplerinin Pilotlar Üzerinde Oluşturduğu Ergonomik Risk Yükleri

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Abstract

Work-related musculoskeletal disorders (WMSDs) are health problems that occur in muscles, tendons, ligaments, joints, and nerves as a result of exposure to ergonomic risk factors in the workplace. Ergonomic risk burden is a measure of a worker's risk of developing musculoskeletal disorders (MSDs) at work. This risk arises from exposure to ergonomic risk factors. Like many occupational groups, pilots are likely to be exposed to similar threats. They work in ergonomically demanding conditions such as prolonged static postures, vibration, and repetitive movements and are exposed to various risks. This increases the risk of developing musculoskeletal disorders. WMSD is one of the leading causes of unemployment and disability among pilots and can lead to long-term health problems. The Quick Exposure Check (QEC) method is a tool that can be used to quickly and easily assess pilots' exposure to these risk factors. This paper presents an application of the Quick Exposure Check method on pilots in the aviation field. It was assessed that different aircraft types may create different ergonomic risk burdens on pilots and the body parts at risk from ergonomic exposure were determined as back, shoulder/arm, wrist/hand and neck.

Keywords: Aviation, Ergonomics, Quick Exposure Check, Ergonomic Risk

Öz

İşe bağlı kas-iskelet sistemi rahatsızlıkları (KİSR), işyerinde ergonomik risk faktörlerine maruz kalma sonucu kaslar, tendonlar, bağlar, eklemler ve sinirlerde oluşan sağlık sorunlarıdır. Ergonomik risk yükü ise bir çalışanın işyerinde KİSR geliştirme riskini gösteren bir ölçüdür. Bu risk, ergonomik risk faktörlerine maruziyetten kaynaklanır. Birçok meslek grubu gibi pilotlar da benzer tehditlere maruz kalma ihtimaline sahiptir. Uzun süreli statik postürler, titreşim ve tekrarlayan hareketler gibi ergonomik açıdan zorlu koşullarda çalışan ve çeşitli risklere maruz kalan bir meslek grubudur. Bu durum, KİSR geliştirme riskini arttırmaktadır. KİSR, pilotlar arasında işsizlik ve sakatlanmanın önde gelen nedenlerinden biridir ve uzun vadeli sağlık sorunlarına yol açabilir. Hızlı Maruziyet Değerlendirme (HMD) yöntemi, pilotların bu risk unsurlarına maruz kalma durumunu hızlı ve kolay bir şekilde değerlendirmek için kullanılabilecek bir araçtır. Bu makalede havacılık alanında Hızlı Maruziyet Değerlendirme yönteminin pilotlar üzerine bir uygulaması yapılmıştır. Birbirinden farklı uçak tiplerinin pilotlar üzerinde farklı ergonomik risk yükleri oluşturabileceği değerlendirilmiş ve ergonomik maruziyetten doğan riskli vücut bölgeleri bel, omuz/kol, bilek/el ve boyun olarak belirlenmiştir.

Anahtar Kelimeler: Havacılık, Ergonomi, Hızlı Maruziyet Değerlendirme, Ergonomik Risk

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1. INTRODUCTION

The intense work pace and complex work environments brought about by modern life can create significant stress and fatigue for employees. This can lead to many problems, such as occupational accidents, work-related musculoskeletal disorders (WMSDs), productivity losses, and decreased product or service quality. Ergonomics science comes into play at this point, tries to protect against these risks, and focuses on optimising the harmony between man and machine. One of these ergonomic risks, WMSDs, covers a range of conditions that affect various components of the musculoskeletal system such as muscles, tendons, joints and nerves. These conditions can lead to pain, fatigue, limitation of movement and even incapacity for work, significantly reducing the quality of life and productivity of workers. Muscle and joint pain, especially in the neck, back, hands and wrists, muscle fatigue and weakness, decreased range of motion, numbness and tingling sensation in the hands and arms, weakness and weakness in the muscles are examples of symptoms of work-related musculoskeletal disorders. Factors that may cause work-related musculoskeletal disorders include non-ergonomic working conditions, heavy load carrying, vibration, repetitive movements, psychosocial factors [1].

The Quick Exposure Check (QEC) Method, which is one of the methods frequently used in ergonomics to analyze the risk of developing work-related musculoskeletal disorders, is a method that enables easy identification of ergonomic risks in the workplace. This method helps to analyze the risk of developing WMSDs by identifying risk factors in the work environment through observation and survey. The QEC method is not a comprehensive ergonomic assessment method. It only helps to predict the risk of developing WMSDs by identifying key risk factors. For a more detailed assessment, more comprehensive methods may need to be used [1, 2].

When the literature is analyzed for the assessment of the risks leading to WMSDs, it is seen that many methods have been developed until today. These methods are divided into three groups: i. recording methods, ii. systematic observational methods, and iii. direct measurement methods according to their structures. In addition, these approaches are divided into two groups: i. reactive and ii. proactive according to their intended use. The proactive approach involves the assessment of risks at the design stage of the system before they arise. The reactive approach, on the other hand, involves the examination of situations such as discomfort and illness that arise as a result of the realization of risks after the implementation of the system and the evaluation of such events to prevent recurrence. While the recording method is used within the scope of the reactive approach, systematic observational methods and direct measurement methods are within the proactive approach. However, in addition to the record methods, the other two methods can also be used for reactive purposes. The record methods is a method frequently preferred by practitioners since it does not require any special training other than basic statistical knowledge for the evaluation of the results. For this purpose, infirmary records kept by enterprises and/or self-reports of employees can be used [3]. Nordic Musculoskeletal Questionnaire (NMQ) [4], Dutch Musculoskeletal Discomfort Questionnaire [5] were used to collect self-reports, Cornell Musculoskeletal

Discomfort Survey [6], Body Discomfort Map [7], Rating of Perceived Exertion (RPE) [8] etc. There are many survey studies. When systematic observational methods are considered, American Conference of Industrial Hygienists Lifting TLV (ACGIH TLV) [9], Revised National Institute for Occupational Health and Safety (NIOSH) Lifting Equation [10], Snook Tables [11], Rapid Upper Limb Assessment (RULA) [12], The Strain Index (SI) [13], Occupational Repetitive Actions Index(OCRA) [14], Quick Exposure Check (QEC) [15], Rapid Entire Body Assessment (REBA) [16] have been developed which require training before risk assessment application. Among these methods, a Turkish guidance document of the QEC method was prepared in 2007 with the support of the Ministry of Labour and Social Security [2].

In this study, the main reasons for choosing the QEC method for the determination of exposures causing work-related musculoskeletal disorders are that this method can be applied with a simple training compared to other methods, it is quick, practical and does not take much time, it is flexible in terms of adapting the questions used in the method to different sectors and does not require high costs.

WMSD is an important problem encountered in almost all sectors and work areas that require physical labour, including the aviation sector. Contrary to popular belief, it does not only cover jobs that require physical labour. Office workers or operations requiring sitting work are also frequently encountered with WMSD. Based on these considerations, when aviation ergonomics is analysed, it is thought that pilots working in aviation are exposed to work-related musculoskeletal disorders. As a matter of fact, the information shared in section 2 of this study also supports the mentioned thoughts.

Aviation ergonomics is a systematic approach to optimizing the performance and safety of all individuals involved in flight operations, such as pilots, cabin crew, and maintenance personnel. As pilots face long flight times, high stress levels, and complex task loads, the application of ergonomic design principles is vital. Ergonomic cockpit design reduces fatigue and physical discomfort by increasing pilots' physical comfort through optimal placement of control panels and displays, seat adjustability and appropriate body supports. Furthermore, improving human-machine interfaces enables more efficient management of information flow, reducing pilots' cognitive load and reducing the likelihood of error. These ergonomic practices allow pilots to perform at a higher-level during flight and make significant contributions to overall flight safety [17].

The aim of this study is to measure the risk burdens on pilots working in the aviation sector with the Quick Exposure Check method and to determine the risk burdens of aircraft types on pilots. It has been determined that work-related musculoskeletal disorders occur in pilots as well as in many occupational groups. It was evaluated that different aircraft types may create different ergonomic risk burdens for pilots. The main reason for this idea is that the design, size and usage characteristics of each aircraft vary with the change of aircraft type. These differences affect factors such as pilots' workload, body postures, access to controls and long and short-term sitting positions. Risky body parts arising from ergonomic exposure were also identified through the application.

This paper is structured as follows: Section 2 provides a literature review, while Section 3 describes the characteristics of the Quick Exposure Check method and how it works. Section 4 describes the application part of the QEC method with the efforts to make it suitable for the aviation sector and pilots. Section 5 evaluates the results obtained by using the QEC method according to the ergonomic risk burdens on pilots and various aircraft types. Section 6 is the part where the results are discussed.

2. LITERATURE REVIEW

In the aviation industry, ergonomics is considered as an important element in terms of employee health and operational efficiency, and it is thought that a comprehensive examination of the risks in this area can contribute to the sustainability of flight safety and performance. Since the scope of this study is pilots working in the field of aviation and the ergonomic risk burdens on these employees, the prominent ergonomic studies in the field of aviation are analysed in this section. It was tried to determine what kind of applications were made in the field of ergonomics in aviation and which methods were used for what purposes.

2.1. Ergonomics Studies in Aviation

The studies conducted in the aviation sector are analysed under three sections as "The Effect of Working Conditions on Neck, Back and Shoulder Pain of Pilots", "Cockpit Design and Ergonomics" and "Anthropometry and Physical Suitability in Pilots".

2.1.1. The Effect of Working Conditions on Neck, Back and Shoulder Pain of Pilots

Neck, back and shoulder pain is an important problem especially for pilots exposed to high G forces. In this context, the study conducted by Sovelius et al. [30] investigated the effects of neck and back muscle loading and the use of lumbar support in pilots exposed to high G forces. Eleven Finnish Air Force pilots participating in the study performed the same air combat manoeuvres with and without lumbar support. Electromyography (EMG) activity measurements showed generally lower %MVC (maximum voluntary contraction) values when the lumbar support was used, but these differences were not statistically significant. Nevertheless, the lumbar support is considered to have the potential to reduce muscle fatigue and improve ergonomic sitting posture. However, individual differences observed between pilots have been noted. These findings suggest that the use of lumbar support may be influenced by personal preferences and physical differences. Another important study on neck and back pain was conducted by Grossman et al. [22]. In this study, back pain symptoms were evaluated among 566 airmen serving in the Israeli Air Force and using different types of aircraft. Based on questionnaire data, the study revealed that back pain was more prevalent among helicopter pilots (89.38%) and attack helicopter pilots (74.55%). In addition, neck pain was more common in fighter and utility helicopter pilots, while low back and mid back pain was reported to be more common and severe in helicopter pilots. These results clearly indicate that back and neck pain may vary by aircraft type. This study demonstrates that aviators face different physical challenges depending on the type of task and emphasises that such challenges may have an impact on pain profiles. Similarly,

another study conducted by Truszczyńska et al. [21] investigated the prevalence and consequences of back pain among Polish Air Force pilots. The study was conducted through a retrospective and anonymous questionnaire on 94 pilots, and information on age, aircraft type, flight hours, pain characteristics, and physical activity levels were collected. Pain levels were determined using visual analogue scales (VAS). The results showed that 59.57% of the pilots experienced back pain. These pains were found to have negative effects on the flight performance and flight safety of the pilots. Therefore, it is clearly seen that the pain experienced during flight affects the fulfilment of the mission and flight safety, therefore, it is emphasised that further research and prevention strategies should be developed. In the study published by Shiri et al. [23], the relationship between cervical (neck) and lumbar (lower back) pain and radiological degeneration in fighter pilots was addressed by a systematic review and meta-analysis method. The study revealed that the prevalence of neck pain increased in pilots exposed to high G-forces, but did not present any different findings in terms of cervical or lumbar disc degeneration. These findings point to the necessity of taking measures to reduce the physical stress of pilots. These results suggest that physical stress may accumulate over time and cause pain and that such stress factors should be more closely analysed. However, in a study conducted by Novais-Shimano et al. [28], the shoulder muscle load of Brazilian Air Force Aerobatic Team pilots was evaluated. In the study, the activation levels of the pilots' shoulder muscles were analysed using surface electromyography (SEMG) and measurements were made under normal and high load conditions. The findings showed that the perception of discomfort increased and muscle activation levels increased in the right shoulder, especially under +G force. The study suggests that physical training programmes that strengthen the shoulder muscles of aerobatic pilots should be revised to minimise discomfort symptoms during flight. Finally, Murray et al. [18] investigated neck and shoulder muscle activity and postures among helicopter pilots and crew members during military helicopter flights. In this study, the physical workload on the neck and shoulder muscles was determined using electromyography (EMG) measurements, and the levels of muscle loads that pilots were exposed to were detailed. These findings once again demonstrate the necessity of ergonomic improvements for helicopter pilots and crew.

2.1.2. Cockpit Design and Ergonomics

Cockpit design and ergonomics are critical to pilot performance and flight safety. One of the important studies in this field was carried out by Lüdtke and Osterloh [31]. The researchers developed a method to support the ergonomic design of aircraft cockpits through the simulation of pilots' cognitive processes. Within the scope of the study, an executable cognitive pilot model was used to estimate the probability of pilots missing information in the displays. This model focuses on how pilots mentally process the current situation and their attentional processes. The model was integrated with formal cockpit designs to simulate human-machine interaction and analysed specifically for the perception of automatic flight mode changes. In addition, how psychological phenomena such as ‘Learned Carelessness’ and ‘Selective Attention’ can lead to pilot errors were also analysed. This study provides valuable data on how models developed on human-

machine interaction can be integrated into cockpit design. In addition, another study by Lijing et al. [32] introduced a method for evaluating the ergonomic design of aircraft cockpits in a virtual environment. The study evaluates the ergonomic layout of the cockpit and the human-computer interface through simulation. This process consists of three stages: importing a digital model of the cockpit into the CEA (Cockpit Ergonomics Layout and Assessment) system and evaluating pilot mannequins with various posture scenarios. The results show that this method can be used in cockpit design to reduce the cost of prototype testing and accelerate the design process. Thus, the importance of digital simulations in cockpit design is revealed once again. Another study by Karmakar et al. [20] examines how digital human modelling and simulation is used in the visual analysis of a jet aircraft pilot. In this study, three dynamic digital pilot models representing the Indian pilot population were created and integrated with a digital prototype of the cockpit. The visual requirements of the pilot were assessed using various visual analysis tools such as fields of view, blind spots, reflection areas, etc. The results show that digital human modelling software is effective in evaluating the layout and alignment of displays and control mechanisms before the physical prototype is developed. This study reveals that digital modelling and simulation can make significant contributions to cockpit design processes. Another study conducted by Monteiro et al. [27] is an ergonomic study on the occupational safety of Airbus pilots in an airline company in Brazil. This study consisted of three main parts and examined the critical situations faced by pilots, the difficulties in the working environment and the effects of irregular working hours on the biological clock. In the research, structured interviews with pilots and systematic observations were conducted using the 'cognitive task analysis' method. In addition, analyses were made by reviewing flight manuals and existing civil aviation, occupational health and ergonomics literatures. This research provides important data on the effects on pilots' occupational safety and biological clock balance.

2.1.3. Anthropometry and Physical Suitability in Pilots

Anthropometric characteristics and physical fitness of pilots are extremely important for ergonomic cockpit design and flight safety. In a study published by Orsello et al. [29], the relationship between low back pain and height among US Navy helicopter pilots was analysed. The study was based on a survey of 1028 helicopter pilots. The findings showed that pilots above a certain height were more likely to experience low back pain. The methods used in the research included case-control analysis, logistic regression and various statistical tests (ANOVA, Chi-square test). The results suggest that pilots' height increases the risk of low back pain and this may be related to ergonomic stressors. These results indicate that cockpit design should be optimised in accordance with the physical characteristics of pilots. Similarly, another study conducted by Şenol [19] examines cockpit ergonomics and assesses critical access and visual capabilities for 90% of potential pilots. The study identifies the discrepancies between pilots' height measurements and cockpit dimensions and provides recommendations to improve ergonomic compatibility. In addition, regression models were developed to ensure external vision adequacy and strong relationships were found between height, eye level

and visual angle. This study provides important data detailing the effects of ergonomic adjustments on pilot performance. In a study published by Da Silva [24] et al., the anthropometric characteristics of Brazilian and US military personnel were compared. The research evaluates the suitability of the US military anthropometry databases applicable to the Brazilian Air Force. In the study, the body measurements of Brazilian pilots and US military personnel were compared using F-test, t-test and percentile 5-95 ranges. The results show that there are significant differences between the two groups and that the US databases are not suitable for use in ergonomic cockpit design for Brazilian pilots. This study highlights the importance of collecting anthropometric data in a way that is representative of the target population. Lee et al. [26] conducted a study on the ergonomic evaluation of pilot oxygen masks designed for Korean Air Force pilots. In this study, the subjective discomfort levels of an existing oxygen mask and a redesigned prototype mask on Korean Air Force pilots, face contact pressure and mask slip distances in high gravity conditions were compared. The research used methods such as questionnaires on pilots, pressure indicating films used to measure facial contact pressure and evaluation of the performance of the mask in high-G conditions. The results show that the redesigned mask offers lower discomfort levels and facial contact pressure than the existing mask.

2.2. QEC Studies in Other Sectors

In the literature review, studies in the field of aviation were examined, but no application of the QEC method, which is widely used in other fields, on pilots was found. Examples of QEC applications in sectors other than aviation are as follows;

Motamedzade et al. [34] compared ergonomic risk assessments of worker positions in a motor oil company in the automotive industry using Rapid Entire Body Assessment (REBA) and Quick Exposure Check (QEC) methods. The study aimed to measure the consistency between the ergonomic risk assessment outputs of the two methods by examining 40 different work postures in a motor oil manufacturing company. It was found that there was a high correlation between the risk assessment results of both methods and that they provided similar risk levels. As a result, it is suggested that researchers can use these methods alternately for postural risk assessment in workplaces. However, the lack of injury records in the dataset is an important limitation as the relationship between the results of the methods and injury data cannot be fully analysed. Similarly, using QEC and REBA methods, Nadri et al. [38] compared the ergonomic risk assessments of workers in a factory in Tehran, Iran using the Quick Exposure Evaluation (QEC) and Rapid Entire Body Assessment (REBA) methods. The study reveals that there is a significant discrepancy between these two methods and therefore they cannot be used interchangeably in work posture risk assessment. As a result of the study, 63.6% of the postures assessed with QEC were included in the high risk group, while only 12.7% were included in the high risk group with REBA. The findings show that there is no significant correlation between the results of QEC and REBA and that there is no statistically significant relationship between musculoskeletal disorders and the results of these methods. The article concludes that these two methods cannot be used interchangeably. This study emphasises that QEC and REBA

may give different results and shows that care should be taken in the choice of method in ergonomic risk assessment. However, it is thought that the limited sample size may reduce the generalisability of the results. Using QEC and REBA methods in the same way as these studies, Pal and Dhara [40] examined work-related musculoskeletal disorders and postural stress in female farmers engaged in uprooting work in rice agriculture. QEC and REBA methods were used and compared in the study. The study indicates that the prolonged and repetitive physical activities of women farmers, especially the demanding body positions they take during uprooting, lead to musculoskeletal disorders and high levels of postural stress. The study emphasises the importance of ergonomic interventions and training for the management and alleviation of such workloads. According to the QEC results, the majority of female workers are at high risk, and the REBA results similarly showed that the posture of the workers needs ergonomic interventions. It is thought that there may be limitations in how suitable the methods used are for continuously mobile and demanding work conditions.

Qutubuddin et al. [37] used the RULA method in addition to the QEC and REBA methods used in previous studies. It deals with an ergonomic risk assessment to analyse the postures of workers working in bus body manufacturing. The study used posture analysis tools such as Rapid Upper Body Assessment (RULA), Rapid Entire Body Assessment (REBA) and Quick Exposure Check (QEC) to assess the ergonomic risks to which workers were exposed. The findings showed that a significant proportion of workers were at moderate to high levels of ergonomic risk and suggested the need to implement ergonomic intervention programmes. The article recommends increasing awareness and education among workers. The study shows how industrial workstations can be improved from an ergonomic point of view. However, since the analysis is limited to a specific industry, it is thought that it may be difficult to generalise the results to other sectors. Using four different methods by adding OWAS method to Qutubuddin et al. study, Saliha et al. [39] evaluates the working postures of female workers in forest nurseries in Turkey. The study analysed workers' working postures and compared their risk scores using various ergonomic risk assessment methods such as REBA, RULA, QEC and OWAS. The findings showed that activities such as working with machinery, manual weeding and carrying were found to have low risk except for the OWAS method. The paper suggests that the RULA method, which is more sensitive, especially for the upper limbs, is more suitable for use in forest nursery work. According to the QEC method, all jobs were found to be high risk, while the REBA and RULA methods found moderate risk in some jobs. The OWAS method gave low risk results for certain jobs. The results show that QEC in particular is more stringent in assessing workers' postures, whereas the other methods are more specific to body parts. The study shows that different ergonomic risk assessment methods may give different results in the same work conditions. However, QEC results based on workers' subjective perceptions may have identified excessively high risks in some cases.

Joseph et al. [35] evaluated the measurement consistency between job analysis methods. The study used three different methods (QEC, OCRA and 4D Watbak) to measure the impact of changes to a workstation in a food processing plant on musculoskeletal (MSD)

risk factors. The results show a positive relationship between the methods, especially between the QEC and OCRA indices. QEC assessed risk factors at regular intervals, while 4D Watbak analysed the worker's postures in more detail. The results showed that QEC gave more consistent results compared to other methods, especially in terms of the back index. In addition, QEC was found to be easier to use than other methods. Chiasson et al. [36], who examined more methods than these studies, compared eight different methods to assess risk factors associated with musculoskeletal disorders in the workplace. The methods include Quick Exposure Check (QEC), Finnish Occupational Health Institute's Ergonomic Workplace Analysis, ACGIH Hand Activity Level, JSI, OCRA index, EN 1005-3 standard, Rapid Upper Body Assessment (RULA) and Rapid Entire Body Assessment (REBA). The study included 224 workstations and 567 tasks from 18 different industrial sectors. The findings show that the methods evaluate the same workstations in different ways. The findings showed that the QEC method was the least stringent, but was 57% similar to OCRA, particularly in assessing hand and wrist risks. QEC identified the lowest level of risk throughout the body, while EN 1005-3 identified the highest risk rates in the shoulder region. The highest correlation was observed between RULA and REBA methods. This study provides information to assist practitioners when selecting methods before planning ergonomic interventions in industry. The study compares many different methods and shows that QEC works in harmony with other methods.

Unlike previous studies in the literature, this study aims to identify the ergonomic risk loads experienced by pilots working in the aviation industry using the Quick Exposure Check (QEC) method, categorized by aircraft type. No studies have been found in the literature that use the QEC method on pilots. Additionally, the advantages of the QEC method over other methods are considered to include rapid and practical assessment, employee participation, identification of stress and strain factors, and the ability to differentiate between dynamic and static work. Considering the necessity and benefits of applying the QEC method to pilots, it can be said that pilots are exposed to risk factors such as staying in a fixed position for long periods during flight, experiencing difficulties in accessing control devices, and encountering excessively repetitive movements. Such ergonomic challenges can lead to musculoskeletal disorders over time and negatively impact pilots' job performance.

Studies conducted under the heading "Studies on Ergonomics in Aviation" have contributed significantly to developing a comprehensive understanding of the physical and cognitive needs of pilots. While research on physical problems such as neck and back pain has achieved limited success in reducing muscle fatigue through solutions like lumbar support, the development of models that optimize human-machine interaction in cognitive ergonomics and cockpit design has contributed to design processes by simulating pilot errors and attention deficits. Additionally, it has been concluded that ergonomic arrangements should be more individualized and efficient by considering factors such as pilot height, flight hours, and aircraft type, which affect health and performance. These studies demonstrate that aviation ergonomics is evolving towards a more holistic and flexible approach in terms of both safety and efficiency.

3. METHODOLOGY

In this study, QEC method was used to identify and evaluate ergonomic risk burdens. Musculoskeletal disorders (MSDs) in workplaces are an important public health problem leading to labour loss and health problems. Ergonomic risk assessment studies are critical in the prevention of these diseases. The Quick Exposure Check (QEC) method is a simple and useful tool that can be used to assess ergonomic risks quickly and practically [1, 2].

The QEC method was developed by Li and Buckle in 1998. The method focuses on assessing ergonomic risk factors that can be observed in the workplace and that employees can report. QEC was first developed in Canada and then started to be used in the UK and other countries [41].

The QEC method consists of two main parts [1]. Table 1 shows these parts and the ergonomic risk factors analysed.

Table 1. QEC method sections and ergonomic risk factors analysed.

Observer's Section	Worker's Section
In this section, ergonomic risk factors that can be observed by the observer in the workplace are assessed.	In this section, ergonomic risk factors reported by the workers are assessed.
• Back position • Back movement/frequency of movement • Shoulder-arm position • Shoulder-arm movement/frequency of movement • Wrist-hand position • Wrist-hand movement/frequency of movement • Neck position	• Weight lifted/carried • Time spent on the task • Applied force level • Visual demand • Driving • Vibration • Work pace • Stress

Risk factors in both sections are scored. The scores obtained from observation and questionnaire results are summed according to body parts (back, shoulder-arm, wrist-hand, neck) and the total score is used to determine the ergonomic risk level. Low scores indicate a low ergonomic risk level and high scores indicate a high ergonomic risk level.

The main stages of the rapid exposure assessment method are as follows; observer selection and training, creating an observation plan, observation, scoring, questionnaire preparation, questionnaire application, scoring, result evaluation, and taking precautions. In Step 1, an observer trained and experienced in ergonomic risk assessment should be selected. The observer should receive comprehensive training on the principles of the QEC method, checklist and scoring system. In Step 2, the workplace and workstations to be assessed should be determined. When and for how long the observations will be made should be planned. Necessary equipment (checklist, pen, notepad, etc.) should be prepared. In Step 3, the observer should observe the workplace and workstations in line with the predetermined plan. The observer should pay attention to the items mentioned above. Record the observations on the checklist. For

each observation, the scoring system specified in the checklist should be used. In step 4, the scores are totalled to obtain the observer score. A questionnaire containing questions about ergonomic risks should be prepared. The questionnaire should be written in simple and clear language that employees can understand. The questionnaire should include a scoring system for each ergonomic risk factor. The questionnaire can be administered face-to-face or online to employees in the workplace [2]. Employees should be given enough time to complete the questionnaire. It should be shared with the employee that the questionnaires will be kept confidential and will only be used for evaluation. In Step 5, the answers given by the employees in the questionnaires should be scored. The employee score is obtained by adding the scores. In Step 6, the observer score and the employee score are summed. The total score is used to determine the ergonomic risk level. Low scores indicate a low ergonomic risk level and high scores indicate a high ergonomic risk level. The necessary measures should be taken according to the results obtained in Step 7. Measures should be aimed at eliminating or minimizing ergonomic risk factors [1, 2].

In conclusion, the QEC method is a useful tool that can be used to quickly and practically assess ergonomic risks in workplaces. The simplicity, usefulness and low cost of the method make it usable in different workplaces. It can be used as part of ergonomic risk assessment programs and should be used in conjunction with expert assessment [2].

4. IMPLEMENTATION

In this study, QEC method was used to determine and evaluate the ergonomic risk burdens of aircraft types on pilots. The applied method consists of 'Determination of Aircraft Types', 'Customisation of QEC Method Questions', 'Data Collection', 'Scoring' and 'Output' sub-steps. Firstly, aircraft types were determined. The 15 questions of the method were updated in order to make them suitable for the piloting profession. Afterwards, the appropriateness of the questions was evaluated with the answers and opinions of expert pilots. It was determined that the answers given to the questions overlapped, that is, the questions were appropriate. Afterwards, the questions were directed to pilots separated according to aircraft types and the answers received were evaluated according to the scoring system of the QEC method and ergonomic risk loads were evaluated. The application steps and details of the method are as follows;

4.1. Determination of Aircraft Types

The study aims to determine the ergonomic risk loads that pilot are exposed to. These risk loads are analyzed separately according to aircraft types. It is considered that each aircraft type may have different ergonomic risk loads. Based on these objectives and estimates, various aircraft types to be studied were determined. If it is necessary to divide it into classes, it is analyzed in 3 sections as jet aircraft, cargo aircraft, and helicopters. The aircraft types are General Dynamics F-16 Fighting Falcon, McDonnell Douglas F-4 Phantom II, CASA CN 235, Airbus A400M Atlas, Eurocopter AS532 Cougar.

4.2. Customisation of the QEC Method Questions

Firstly, the questions used in the QEC method were analyzed. As a result of this examination, it was determined that the comprehensibility of the questions was difficult for the pilots. In order to make it easier to understand, some linguistic changes were made in the questions and they were made appropriate. The process of adapting to the aforementioned questions was carried out with the support of experienced pilots.

4.3. Data Collection

The questionnaire method was used for data collection. The purpose and objectives of the questionnaire were clearly defined, and it was determined how the data to be collected would be used. The target group consists of pilots. Mainly retired and experienced pilots with high flight times were preferred. Pilots with flight experience in previously selected aircraft types were identified and trained on the HMD method and the purpose of the study. The employee evaluation part of the updated QEC method questions was directed to the pilots identified as the target group. The questions of the observer evaluation section were answered by watching the video recordings recorded from inside the cockpit and analysed in detail by the observer, as well as tests and examinations in simulation cockpits.

4.4. Scoring

At this stage, the answers received from the pilots were processed into the scoring tables of the QEC method and scores were obtained from the matrices according to the body exposure zones. For example, a question is asked in the back section for F-16 aircraft and multiple-choice answers are given below.

A sample question from the observer assessment section (Back Posture);

Back during flight:

A1 In an almost neutral (straight) position?
A2 Moderately tilted forwards or sideways or turned sideways?
A3 Is it excessively tilted forwards or sideways or turned sideways?

A sample question from the pilot assessment section (Weight);
- What is the maximum weight that you have lifted and/or carried by hand while performing this task? (The weight will be expressed as an estimate of the magnitude felt.) ("X" represents the weight lifted.)

H1 $X \leq 5$ kg
H2 $5 \text{ kg} < X \leq 10$ kg
H3 $10 \text{ kg} < X \leq 20$ kg
H4 $X > 20$ kgs

The above two questions analysing back posture (A) and weight (H) were answered by the observer and the pilot. For the F-16 aircraft, the observer chose A1 and the pilot chose H1. Therefore, the score value at the A1/H1 intersection was analyzed from the back posture and weight matrix given in **Figure 1**. The score of the mentioned A1/H1 intersection is 2. The score value obtained from this matrix was noted in the 'Score 1'

box. Like the example given for the back, all matrices were calculated and the score of the relevant section was obtained by summing the exposure scores section by section. It is calculated with 4 matrices in the back section, 5 in the shoulder/arm section, 5 in the wrist/hand section, 2 in the neck section, 1 in the driving section, 1 in the vibration section, 1 in the work tempo section and 1 in the stress section.

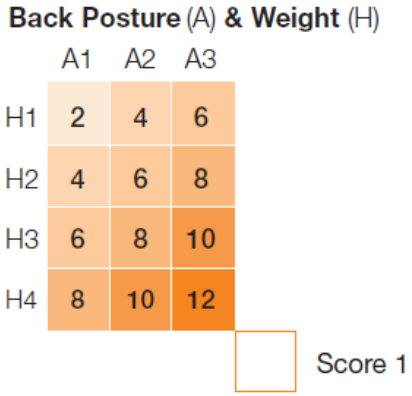


Figure 1. Back Posture (A) and Weight (H) matrix of the back section in the QEC method [1].

After all matrices are calculated and summed, the scores of the exposure areas will be included in one of the 'low', 'medium', 'high', 'very high' exposure levels shown in Table 2.

Table 2. Exposure score/Exposure level table.

Score	Low	Medium	High	Very High
Back (Static)	8-15	16-22	23-29	30-40
Back (Moving)	10-20	21-30	31-40	41-56
Shoulder/Arm	10-20	21-30	31-40	41-56
Wrist/Hand	10-20	21-30	31-40	41-56
Neck	4-6	8-10	12-14	16-18
Driving	1	4	9	-
Vibration	1	4	9	-
Work Pace	1	4	9	-
Stress	1	4	9	16

4.5. Output

After the data collection phase, the answers collected from the pilots and the observer were entered into the QEC evaluation tables. The evaluation results were obtained for back, shoulder-arm, wrist-hand, neck, driving, vibration, work tempo, and stress. The ergonomic risk loads to which pilots were exposed were analyzed in 4 groups as low, medium, high, and very high.

4.6. Results

QEC method was used to determine the ergonomic risk loads on pilots according to different aircraft types. The final results of the application are shown in Table 3, Table 4, and Table 5.

Table 3. Exposure scores and levels for jet aircraft class.

	F-16 Fighting Falcon		F-4 Phantom II	
Exposure Factor	**Score**	**Exposure Level**	**Score**	**Exposure Level**
Back (Static)	20	Medium	24	High
Back (Moving)	-	-	-	-
Shoulder/Arm	22	Medium	28	Medium
Wrist/Hand	22	Medium	34	High
Neck	18	Very High	18	Very High
Driving	4	Medium	4	Medium
Vibration	1	Low	4	Medium
Work Pace	9	High	9	High
Stress	9	High	16	Very High

Table 4. Exposure scores and levels for cargo aircraft class.

	CASA Cn 235		A400M Atlas	
Exposure Factor	**Score**	**Exposure Level**	**Score**	**Exposure Level**
Back (Static)	20	Medium	20	Medium
Back (Moving)	-	-	-	-
Shoulder/Arm	26	Medium	18	Low
Wrist/Hand	28	Medium	22	Medium
Neck	16	High	16	High
Driving	4	Medium	4	Medium
Vibration	1	Low	1	Low
Work Pace	9	High	9	High
Stress	9	High	9	High

Table 5. Exposure scores and levels for helicopter class.

	AS532 Cougar	
Exposure Factor	**Score**	**Exposure Level**
Back (Static)	26	High
Back (Moving)	-	-
Shoulder/Arm	32	High
Wrist/Hand	22	Medium
Neck	16	High
Driving	4	Medium
Vibration	4	Medium
Work Pace	9	High
Stress	16	Very High

When the results obtained are analysed; the scores of the F-16 Fighting Falcon and F-4 Phantom II aircraft examined in the jet aircraft class were close to each other for the back region. However, according to the score/exposure level table of the QEC method, while the exposure level of the F-16 jet aircraft is medium, it poses a high risk for the F-4. In the shoulder/arm region, the situation was reversed in terms of score. However, the exposure level was determined as medium for both jet aircraft. The reason why the F-4 aircraft scored worse is that the control units are not electronic and require more force to move these units. In the wrist/hand section, the situation is similar to the situation in the shoulder/arm section. Since the control units of the F-4 aircraft are not electronic and require more force to move them, it is quite different from the F-16 aircraft. Since the 'fly-by-wire' system is used in the F-16 aircraft, the force required to ensure the movement of the control units is very low. For these reasons, exposure levels in the wrist/hand section were found to be medium for F-16 and high for F-4. The ergonomic risk load in the neck section is the same and very high. The reason is that jet pilots are exposed to high G forces and rotate their necks 90° horizontally and vertically many times during flight. In the driving section, there is moderate exposure for both aircraft. In the vibration section, the F-4 aircraft transmits slightly more vibration. The work tempo is the same and high for both aircraft. Stress is high for pilots of both aircraft. While it is high for F-16 pilots, it is very high for F-4 pilots. It is thought that the reason is that the F-16 aircraft is newer than the F-4 aircraft and offers technological facilities.

CASA CN 235 and A400M Atlas aircraft were analyzed in the cargo aircraft class. In the shoulder/arm area, CASA scored medium and A400M scored low. It is thought that the fact that A400M pilots use their arms at a lower level than CASA pilots play an effective role. For cargo aeroplanes, exposure levels in all other sections except the shoulder/arm section gave the same results. Only the scores in the wrist/hand section differed. While the wrist/hand score of the CASA aircraft was 28, the score of the A400M aircraft was 22. The reason for the score difference in this section is that the control units of the

A400M aircraft are electronically assisted, as in the F-16 aircraft. Although the scores are different, the exposure levels are moderate for both cargo aircraft. The graph in Figure 2 shows the exposure scores according to aircraft types.

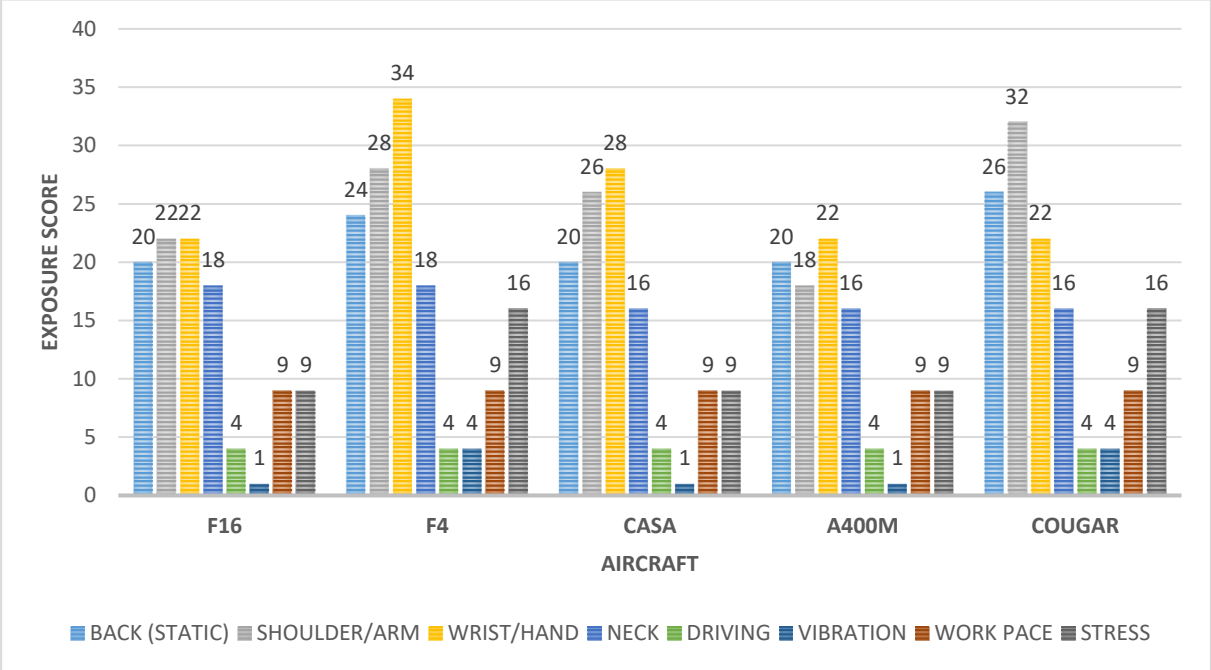


Figure 2. Exposure scores obtained according to aircraft types.

When evaluated among aircraft classes, the helicopter class has the highest exposure in the back region. The closest score to the helicopter's back exposure belongs to the F-4 aircraft. In the shoulder/arm section, the helicopter class has the highest exposure as in the back area. The A400M cargo aircraft has the lowest score. The reason for this is that the arms of the A400M pilots are at a low level. The F-4 jet aircraft has the highest exposure in the wrist/hand region. The reason is that the control units of the F-4 aircraft are technologically outdated and do not have electronic support, in other words, they require force. In F-16 and A400M aircraft, which have the aforementioned electronic support, the wrist/hand exposure level is quite low. The highest exposure in the neck area belongs to jet aircraft. The reason is that jet pilots rotate their heads at higher angles than pilots of other aircraft types due to reaching high speeds during flight and the requirements of the missions. In terms of vehicle use, i.e., flight times and work tempo, the same scores were obtained for all aircraft types. In the vibration section, the F-4 aircraft and the Cougar of the helicopter class received higher scores than the other aircraft. When the stress section was analyzed, the F-4 and Cougar helicopter received the highest scores. For helicopter pilots, the mission difficulty carries this stress to high levels, while for the F-4, both the mission difficulty and the fact that the aircraft is technologically older and has manual features compared to the F-16 aircraft are considered. The graph in Figure 3 shows the exposure scores and exposure levels of the aircraft.

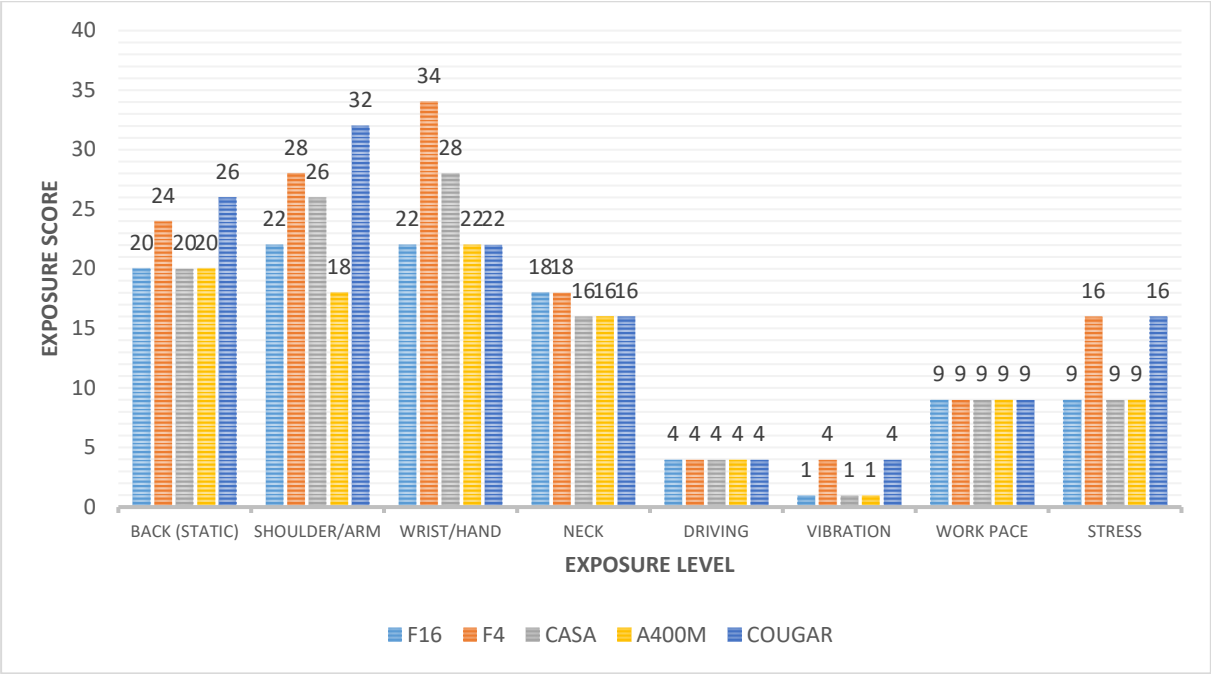


Figure 3. Obtained exposure scores and exposure levels of aircraft.

5. CONCLUSION

Within the scope of this study, an application of the Quick Exposure Check method in the field of aviation was carried out on pilots. It was evaluated that different aircraft types may create different ergonomic risk loads on pilots and risky body parts arising from ergonomic exposure were determined. When the literature was examined, it was seen that studies similar to this study had been conducted on pilots before, but the QEC method was not found. This is the first study in which the QEC method was applied on pilots.

Understanding ergonomic risks in the aviation industry is believed to have significant implications for worker health and operational efficiency. The findings suggest that these risks, if managed correctly, can yield positive results at both individual and organizational levels, potentially improving the performance of pilots and other aviation personnel. Ergonomically improved work processes can support operational sustainability by saving time and costs.

Musculoskeletal disorders, fatigue, and repetitive movements are expected to have negative impacts on pilots' job performance. They can reduce pilots' concentration and increase the likelihood of errors. Fatigue, inattention, and loss of physical control due to musculoskeletal disorders can compromise flight safety.

Reducing these risks can enhance operational efficiency and flight safety by maximizing workers' physical and mental capabilities. As an application, task rotations could be redesigned within the framework of ergonomic assessments. Additionally, occupational diseases, especially musculoskeletal disorders, are considered to lead to serious long-term health problems and workforce losses. Moreover, reducing ergonomic risks can

contribute to workforce retention by increasing employee satisfaction and motivation. Employees feeling more comfortable and motivated in an ergonomically optimized work environment may be less likely to leave their jobs.

In this study, firstly, the questions included in the QEC method were adapted to pilots and the piloting profession. The main reason for reorganizing the questions is that they can be more easily understood by the employees. After the arrangement of the questions, aircraft types were classified as jet, cargo and helicopter. According to this classification, the aircraft to be evaluated were selected. In the next stage, experienced pilots, i.e., pilots with high flight times, were determined to be included in the application. Employees were provided with the necessary training and information about the questions to be asked. In order to ensure the accuracy and appropriateness of the answers given by the pilots and to provide the closest outputs to the reality of the study, the question/answer part was carried out by face-to-face interview method. With the results obtained, the scoring section of the QEC method was applied and the scores of the exposure factors were obtained. In return for these scores, outputs were obtained according to the exposure score/exposure level table given in Table 2.

When the obtained ergonomic risk loads are analyzed according to aircraft types, it is seen that the most important factor is the technology used in the aircraft. It is predicted that ergonomic risk loads will decrease in aircraft with new or modernized technologies. Manual or hydraulic control systems, which are moved by pilots, require the user to apply extra force. For example, it has been observed that this situation is the opposite in electronic control systems, that is, it does not require much force. It was found that the flight times, work pace and the stress experienced by the pilots were generally close to each other in all aircraft types.

When the pilot profession was analyzed in terms of ergonomic risk burdens, it was seen that it has risks like many other occupational groups. It was found that the mentioned ergonomic risk loads vary considerably according to aircraft types. Due to this variability, it has been observed that different aircraft types will have different ergonomic risk loads on pilots. It would be nice if we add the most prominent risks in aircraft together with aircraft variability to this section. It has been observed that the risk group that varies the most with the variability of the plane is the shoulder/arm.

A weak feature of the QEC method is that the method makes a categorical-based assessment. Class distinctions between categories are defined with precise boundaries. However, the application of the QEC questionnaire involves subjectivity and the analysis is expected to be flexible in accordance with this subjectivity. Therefore, in future studies, the QEC method can be developed on a fuzzy set basis and thus more sensitive analyses can be performed.

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